

Quadratic Function Report

Polynomial

$$f(q) = aq^2 + bq + c$$

Naive

$$g_{\text{naive}}(X) = X^2a + Xb + c$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 2a \frac{\Delta^2}{\epsilon} + aq^2 + bq + c$$

$$\text{Var}(g_{\text{naive}}(X)) = 20a^2 \frac{\Delta^4}{\epsilon} + 8a^2 \frac{\Delta^2}{\epsilon} q^2 + 8ab \frac{\Delta^2}{\epsilon} q + 2b^2 \frac{\Delta^2}{\epsilon}$$

$$\text{MSE}(g_{\text{naive}}) = 24a^2 \frac{\Delta^4}{\epsilon} + 8a^2 \frac{\Delta^2}{\epsilon} q^2 + 8ab \frac{\Delta^2}{\epsilon} q + 2b^2 \frac{\Delta^2}{\epsilon}$$

$$\text{Bias}(g_{\text{naive}}) = 2a \frac{\Delta^2}{\epsilon}$$

Unbiased

$$g_{\text{unb}}(X) = X^2a + Xb - 2a \frac{\Delta^2}{\epsilon} + c$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^2 + bq + c$$

$$\text{Var}(g_{\text{unb}}(X)) = 20a^2 \frac{\Delta^4}{\epsilon} + 8a^2 \frac{\Delta^2}{\epsilon} q^2 + 8ab \frac{\Delta^2}{\epsilon} q + 2b^2 \frac{\Delta^2}{\epsilon}$$

$$\text{MSE}(g_{\text{unb}}) = 20a^2 \frac{\Delta^4}{\epsilon} + 8a^2 \frac{\Delta^2}{\epsilon} q^2 + 8ab \frac{\Delta^2}{\epsilon} q + 2b^2 \frac{\Delta^2}{\epsilon}$$

Comparison

$$\text{mean gap} = -2a \frac{\Delta^2}{\epsilon}$$

$$\text{variance gap} = 0$$

$$\text{MSE gap} = -4a^2 \frac{\Delta^4}{\epsilon}$$

$$\text{Bias}(g_{\text{naive}})^2 = 4a^2 \frac{\Delta^4}{\epsilon}$$